

Microwave Properties of Tunable Phase Shifter Using Superconductor/Ferroelectric Thin Films

Min Hwan Kwak,* Young Tae Kim, Seung Eon Moon,
Han-Cheol Ryu, Su-Jae Lee, and Kwang Yong Kang

Basic Research Laboratory, Electronics and Telecommunications Research Institute,
161 Gajeong-Dong, Yuseong-Gu, Daejeon, 305-350, Korea

ABSTRACT

High temperature superconductor, $\text{YBa}_2\text{Cu}_3\text{O}_{7-\delta}$ (YBCO), and ferroelectric, $\text{Ba}_{0.1}\text{Sr}_{0.9}\text{TiO}_3$ (BST), multilayer thin films were deposited on MgO (100) substrates using a pulsed laser deposition. The thin films exhibited only (00 l) peaks of YBCO and BST. The HTS thin films demonstrated excellent zero resistance temperature of 92.5 K. We designed and fabricated HTS ferroelectric phase shifter using high frequency system simulator and standard photolithography method, respectively. The YBCO/BST phase shifter shows a low insertion loss (2.97 dB) and large phase change (162°) with dc bias of 40 V at operating frequency of 10 GHz. The HTS ferroelectric phase shifter shows 54 of figure of merit. These results can be applicable to phased array antenna system for satellite communication services.

Keywords: Ferroelectric and superconducting multilayer thin film; superconducting phase shifter; phase shifter; microwave property; ferroelectric thin film

INTRODUCTION

The advantage of multilayer superconducting and ferroelectric materials is that a low microwave loss could be realized for electrically tunable microwave devices. Ferroelectric strontium titanate (STO) exhibits a non-hysteretic polarization with electric field at liquid nitrogen boiling temperature (77 K) and a lattice constant close to that of $\text{YBa}_2\text{Cu}_3\text{O}_{7-\delta}$ (YBCO) high temperature superconducting (HTS) thin films. With these reasons, STO have been widely used as a substrate for HTS tunable microwave devices such as filters, resonators,

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*Corresponding author. E-mail: mhkwak@etri.re.kr

and phase shifters [1–5]. The permittivity of ferroelectric materials exhibits strong dependence of temperature and electric field. At a specific temperature called ‘Curie temperature,’ the permittivity of ferroelectric materials was rapidly changed. Since STO has so low Curie temperature of ~ 10 K, it exhibits negligible permittivity change under electric field variations at 77 K. For application of phase shifter using the HTS and ferroelectric multilayer thin film, ferroelectric thin films should have larger permittivity changes than that of STO at 77 K.

Ferroelectric $\text{Ba}_x\text{Sr}_{1-x}\text{TiO}_3$ (BST) was attractive material for tunable microwave applications [6–9], since lattice constant, Curie temperature, and permittivity can be controlled by changing the ratio of BaTiO_3 and SrTiO_3 .

In this work, chemical composition of BST with $x = 0.1$ has been selected, because its Curie temperature is close to 77 K [10]. YBCO/BST multilayer thin films were prepared using a pulsed laser deposition method on MgO substrate. Coplanar waveguide (CPW) phase shifters were designed and fabricated using YBCO/BST and Au/BST on MgO substrates to study metal effect of the phase shifters. Microwave properties of the phase shifters such as insertion loss, phase difference with DC bias field from 0 to 40 V have been investigated.

EXPERIMENT

Ferroelectric $\text{Ba}_{0.1}\text{Sr}_{0.9}\text{TiO}_3$ (BST) thin film and HTS YBCO thin film were *in situ* grown on MgO (100) substrates by a pulsed laser deposition. The ablation performed with a KrF excimer laser at a wavelength of 248 nm, a laser fluence of ~ 2.5 J/cm², and a target to substrate distance of 5 cm. The laser repetition rate was 5 Hz during deposition of BST and YBCO film. Single crystal MgO (100) substrate with dimensions $15 \times 15 \times 0.5$ mm³ was mounted on the heater block with silver adhesive. For the uniformity of the thin film, heating block was scanned during deposition.

The BST/YBCO multilayer was fabricated by *in situ* pulsed laser deposition. The BST layer was deposited at oxygen pressure of 180 mTorr and substrate temperature of 790°C. The HTS YBCO layer was deposited with oxygen pressure of 200 mTorr and substrate temperature of 750°C. The heat treatment of the HTS film was performed at 450°C for 1 hour under oxygen pressure of 750 Torr. Thickness of BST and HTS films were measured to be 230 nm and 450 nm, respectively.

The YBCO/BST/MgO films were characterized by $\theta-2\theta$ x-ray diffraction analysis (XRD). A scanning electron microscopy (SEM) was used to examine the surface morphology and cross sectional view of the films.

For comparison with the HTS phase shifter, normal metal phase shifter has been fabricated on BST. For the electrode pattern of normal metal phase shifter, a chromium adhesion layer and a thick gold layer (2 μm) have been deposited on the BST thin films using multi target DC magnetron sputtering system.

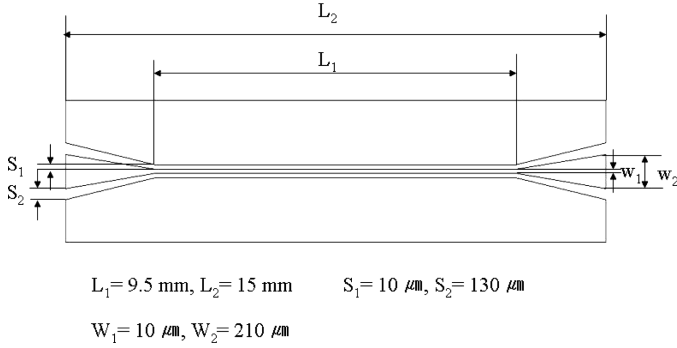


Figure 1. Schematic of the designed and fabricated HTS/BST CPW phase shifter.

A CPW phase shifter was designed by a circuit simulator and an electromagnetic simulator as shown in Fig. 1. Pattern of the phase shifter was fabricated by a standard photolithography technique with a RF ion milling process.

The microwave properties of the phase shifter were measured by a HP 8510C vector network analyzer (VNA) with two ports measurement configuration using metal package in the frequency range from 8 to 12 GHz.

RESULTS AND DISCUSSION

Figure 2 shows a typical θ - 2θ XRD patterns of an YBCO /BST thin film deposited by PLD onto (100) MgO single crystal. The films exhibit highly (00/) oriented YBCO and BST films. Highly oriented (00/) YBCO peaks suggest that its lattice parameter match well with that of BST ($x = 0.1$) film.

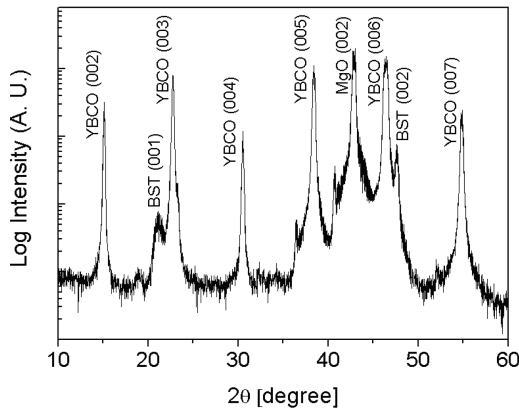


Figure 2. XRD pattern of YBCO/BST/MgO multilayer thin film.

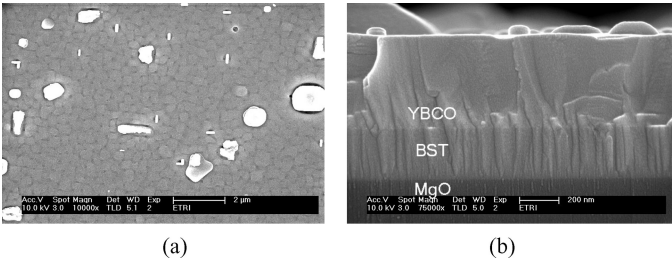


Figure 3. (a) Surface and (b) cross sectional SEM micrograph of YBCO/BST/MgO thin film.

Figure 3 shows (a) surface and (b) cross-sectional view of the YBCO/BST multilayer thin film observed by a scanning electron microscope. The surface of the film was rough with extrusions. However, it has been known that the HTS film with such morphology usually exhibit a large critical current due to large flux pinning force. This type HTS thin film was widely used in microwave applications. In the cross-sectional view, it is clear that the BST layer was grown in column. The Thickness of YBCO and BST layers are 450 and 230 nm, respectively.

The electrical properties of HTS YBCO were measured by a conventional four point probe method with cryo-cooling system as shown in Fig. 4. The YBCO/BST/MgO multilayer thin films demonstrated excellent zero resistance temperature of 92.5 K.

Au and HTS phase shifters were fabricated on the ferroelectric BST films using the dimension shown in Fig. 1. These phase shifters were mounted in brass packages with K-connectors.

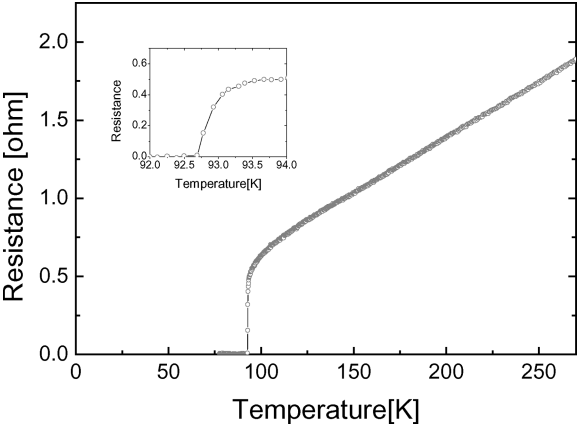


Figure 4. Resistance-Temperature curve of BST/YBCO/MgO thin film.

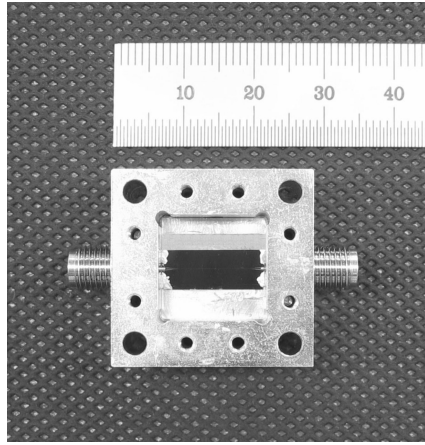


Figure 5. Photograph of packaged HTS/BST/MgO CPW phase shifters.

The low temperature microwave properties were measured by a HP8510C vector network analyzer with cryostat in the temperature range from 300 to 30 K.

The phase shifter loaded packages have been placed on the cold head of the cryostat as shown in Fig. 5. The microwave loss and phase difference of the both phase shifters were measured at 77 K as shown in Figs. 6 and 7, respectively. A range of frequency, 8~12 GHz, was applied on the phase shifters with external bias voltage of 0~40 V.

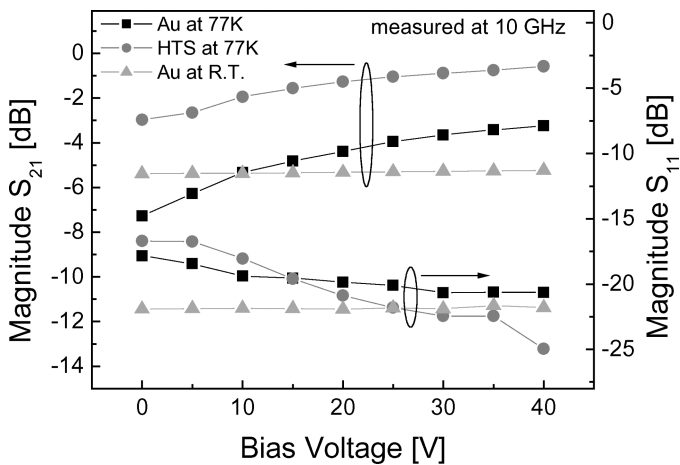


Figure 6. Transmission properties of HTS ferroelectric phase shifter as a function of dc bias voltage at 10 GHz. (See Color Plate XI)

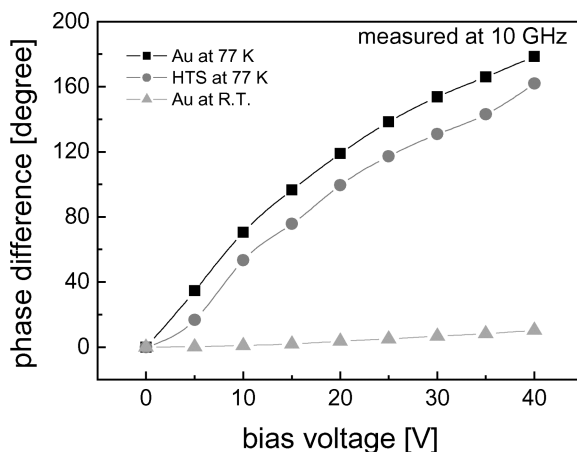


Figure 7. Phase difference of the HTS ferroelectric phase shifter as a function of dc bias voltage at 10 GHz. (See Color Plate XII)

Both phase shifters exhibited low return loss (S_{11}), less than 15 dB, which suggest that the impedance of the CPW match well with 50 ohm.

The YBCO/BST phase shifter shows a low insertion loss (less than 3 dB) and large phase change (162°) with 40 V dc bias at 10 GHz, while the Au phase shifter exhibits a large insertion loss of 7 dB and phase change of 178° . The HTS/BST phase shifter shows 54 of figure of merit, it was about two times than that of Au phase shifter, This result suggest that the HTS phase shifter can be applicable to phased array antenna system for satellite communication services. Furthermore, the conductor loss from the normal metal electrode (Au) of phase shifter contributes a larger part of insertion loss. Another word, electrode loss is larger than dielectric loss of ferroelectrics in CPW type phase shifter.

CONCLUSION

Epitaxial YBCO/BST multilayer was successfully deposited on MgO (100) substrates using a pulsed laser deposition. The thin films exhibited only (00l) peaks of YBCO and BST. The HTS thin films demonstrated excellent zero resistance temperature of 92.5 K due to good lattice match between YBCO and BST. The HTS-ferroelectric phase shifter shows a low insertion loss (2.97 dB) and a large phase change (162°) with 40 V dc bias at 10 GHz. These results can be applicable to HTS phased array antenna system for satellite communication services.

REFERENCES

1. P. Woodall, K. Bouzehouane, B. Marcilhac, D. G. Crete, E. Jacquet, J. C. Mage, and J. P. Contour, *IEEE Trans. Appl. Supercond.* **11**, 1150 (2001).
2. B. H. Moeckly, Luke S.-J. Peng, and G. M. Fischer, *IEEE Trans. Appl. Supercond.* **13**, 712 (2003).
3. D. Galt, J. C. Price, J. A. Beall, and T. E. Harvey, *IEEE Trans. Appl. Supercond.* **5**, 2575 (1995).
4. F. A. Miranda, K. B. Bhasin, K.-S. Kong, T. Itho, and M. A. Stan, *IEEE Microwave Guided Wave Lett.* **2**, 287 (1992).
5. A. T. Findikoglu, Q. X. Jia, and D. W. Reagor, *IEEE Trans. Appl. Supercond.* **7**, 2925 (1997).
6. H.-C. Ryu, S. E. Moon, S. S.-J. Lee, M. H. Kwak, Y. T. Kim, and K.-Y. Kang, *Jpn. J. Appl. Phys.* **43**, 6746 (2004).
7. Y.-T. Kim, H.-C. Ryu, M. H. Kwak, S. E. Moon, S.-J. Lee, S.-H. Kim, J.-S. Park, and S.-K. Han, *Integrated Ferroelectrics* **56**, 1107 (2003).
8. S. E. Moon, E.-K. Kim, M.-H. Kwak, H.-C. Ryu, Y.-T. Kim, K.-Y. Kang, and S.-J. Lee, *Appl. Phys. Lett.* **83**, 2166 (2003).
9. S. W. Kirchoefer, J. M. Pond, H. S. Newman, W.-J. Kim, and J. S. Horwitz, *IEEE MTT-S International Microwave Symp. Dig.* 1359 (2000).
10. H. Kobayashi and T. Kobayashi, *Jpn. J. Appl. Phys.* **33**, L533 (1994).